MATTHEW H. ZHANG

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EDUCATION

The University of Texas at Austin – GPA: 3.52 May 2029
Bachelor of Science in Electrical and Computer Engineering Honors
Bachelor of Business Administration in Canfield Business Honors
Relevant Coursework: Circuit Theory Honors, Circuit Analysis & Design Honors, Linear Systems & Signals, Embedded Systems Honors, Computing Systems Honors, Software Design Honors, Data Science Honors

EXPERIENCE

Electrical Engineering Intern – *eve Vehicles Corporation* Oct 2026 – Present

- Designing battery quick-swap and power distribution board with integrated sensing for autonomous UAVs

Member of Technical Staff – *Pantheon Industries ($25M+ raised, Kleiner Perkins)* Jun 2026 – Aug 2026

- Led tactile-sensing R&D across magnetic, optical, capacitive, and resistive approaches; built piezoresistive thru/shunt-mode FPC (flex PCB) prototypes and a calibration rig from a repurposed 3D printer, load cell, and fiducial vision tracking
- Owned schematic, layout, firmware, and bring-up of a 40x40mm STM32 CAN FD-UART bridge with 24V-to-11V 6A buck
- Deployed CV gripper calibration tools, reducing operator downtime 92% and increasing per-station data capture 110%
- Drove all electrical engineering efforts and assisted mechanical development and fabrication of 150+ data-collection devices
- Schematic capture of 1 kW 24V/50A bimanual PDB + USB-hub with triple buck rails and 5 Gbps data transmission

Sensors Integration Lead – *Texas Humanoids* Sep 2026 – Present

- Leading 4 engineers in tactile sensing R&D for an open-source humanoid with UT's Robotics Lab
- Designing a magnetic + FSR tactile sensor stack for fingertips to measure shear + normal force, read out over CAN

Engineering Lead – *Texas Aerial Robotics* Jan 2026 – Oct 2026

- Led 15 engineers across electrical, mechanical, and ML to explore a 6-DOF omnidirectional gimballed-thrust quadcopter for executing in-flight visuomotor policies trained from UMI demonstrations
- Ran PCB onboarding for new members; reviewed schematics and layouts for DRC, footprint, and power-routing errors
- Designed compact AM32-based 30A 4S BLDC motor driver PCB under size and thermal constraints
- Modeled inverted pendulum and TVC system dynamics and implemented PID, MPC, & LQR controllers in Python

Embedded Systems Engineer – *GHOST Robotics, VEX U* Nov 2025 – May 2026

- Redesigned differential I2C sensor bridge PCB with address translation and reduced board area by 40%
- Wrote interrupt-based RP2040 sensor polling firmware in C with custom sensor drivers over USB serial
- Performed board bring-up and debug by validating power rails and signals with a multimeter and logic analyzer

PROJECTS

BetaGo, Autonomous Chess Robot – 1st Place @ World Tour Physical AI Hackathon May 2026

- Implemented classical, VLM, VLA, and IK control methods to autonomously play chess with a robot arm
- Introduced CNN segmentation and end effector adjustments to increase pick and place accuracy by 61%

Inhabit – *Co-Founder, NVIDIA Inception Startup & Founders Inc Residency* Dec 2025 – Jun 2026

- Led development of a wearable motion capture system streaming human joint angles at 50 Hz over USB serial
- Designed MCU-based magnetic encoder board using RS-485 daisy-chaining, enabling plug-and-play joint modules
- Deployed and validated on Unitree G1 EDU for episodic task collection

SKILLS

Technical Skills: Circuit Design, PCB Layout, Firmware Development, Control Systems, Robot Learning, Computer Vision, CAD, Mechanical Design, Rapid Prototyping, Git, Debugging, Soldering | **Tools:** KiCad, Altium, LTspice, MuJoCo, OpenCV, Linux, Onshape, 3D Printing, GDB, Oscilloscope, Logic Analyzer | **Languages:** C, C++, Python, Assembly, R, SQL | **Embedded:** CAN, I2C, SPI, UART, RS-485, ADC/DAC, BLE

AWARDS

Stanford DroneHacks Rothberg Catalyzer Award, 1st Place (2026) • UC Berkeley AI Hackathon Best Physical AI Hack (2026) • Founders Inc Physical AI Hackathon 1st Place (2026) • Intrinsic AI for Industry Challenge 12th of 160 Teams (2026) • UT x CSBA x HBA Hackathon 1st Place Track (2026) • IEEE RAS Robotathon 1st Place, Software Design Award (2025) • FIRST Robotics Rising All Stars Award (2025) • FIRST Robotics Worlds Qualifier, World's Rookie Inspiration Award (2024)